

Fortinet Report

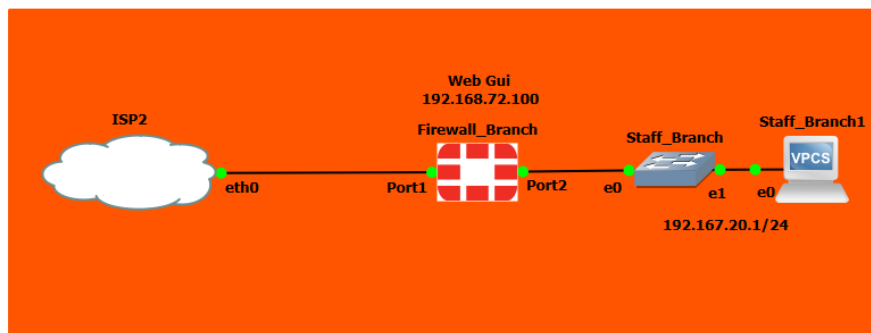
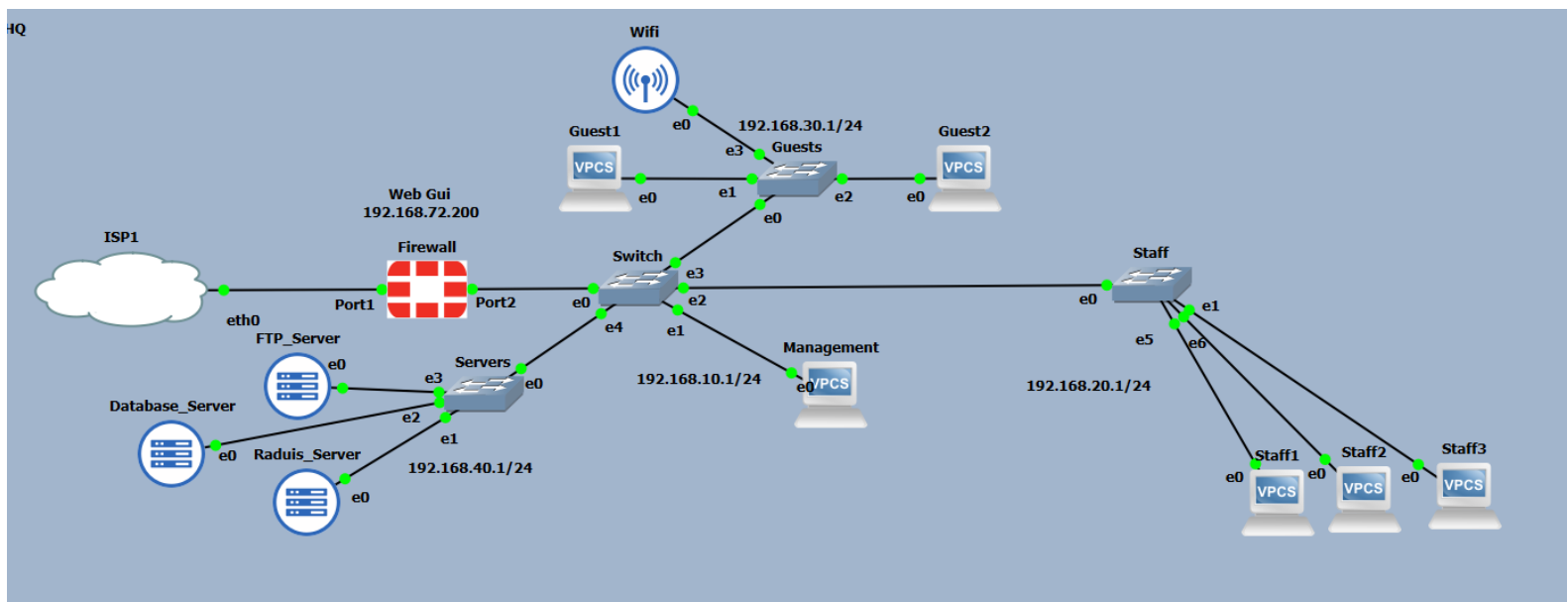
Network Security Fundamentals and FortiGate Integration

#2 adding

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The network topology



Policies after

+ Create New Edit Delete Policy Lookup Search Q Interface Pair View By Sequer											
ID	Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes	
HQ-to-Branch --> Staff (Vlan 20)											
5	vpn_HQ-to-Branch_remote	HQ-to-Branch_remote	HQ-to-Branch_local	always	ALL	ACCEPT	Disabled	default certificate-inspection	All	4.90 kB	
Servers (Vlan 40) --> Staff (Vlan 20)											
7	Guests Internet	all	all	always	DNS FTP MYSQL	ACCEPT	Enabled	default default default default	All	0 B	
Staff (Vlan 20) --> HQ-to-Branch											
4	vpn_HQ-to-Branch_local	HQ-to-Branch_local	HQ-to-Branch_remote	always	ALL	ACCEPT	Disabled	default certificate-inspection	All	0 B	
Staff (Vlan 20) --> Servers (Vlan 40)											
6	Staff To Servers	all	all	always	FTP MYSQL RADIUS	ACCEPT	Enabled	default certificate-inspection	All	0 B	
Staff (Vlan 20) --> WAN (port1)											
3	Staff Internet	all	all	always	ALL	ACCEPT	Enabled	default default default default	All	216 B	
Implicit											
0	Implicit Deny	all	all	always	ALL	DENY			Disabled	2.52 kB	

ID 3: Staff Internet

Purpose:

Allow staff PCs (Vlan 20) to browse the internet via the WAN interface.

Explanation:

This is a standard outbound internet access rule. All services are allowed (HTTP, HTTPS, DNS, etc.), and NAT is enabled so staff devices can use the firewall’s public IP. Security profiles are applied to protect the users from malicious sites and traffic.

Why it’s important:

This is the primary user internet access policy, essential for real-world network functionality.

ID 4: pn_HQ-to-Branch_local

Purpose:

This is the **return traffic rule** for the VPN — enabling the Branch Office to communicate back to the HQ internal LAN.

Explanation:

Traffic coming **from the remote branch** (vpn_HQ-to-Branch_remote) is allowed into the HQ local network (HQ-to-Branch_local). This policy is symmetrical with Policy 5, enabling **two-way communication** between HQ and Branch via VPN. NAT is disabled because VPN communication must use real internal IPs, not translated ones.

Why it’s important:

Without this policy, the VPN tunnel would be **one-way only**, breaking branch-to-HQ connectivity.

ID5: HQ-to-Branch

This policy enables secure communication between the **Headquarters LAN (Staff Vlan 20)** and the **Branch Office network** through a VPN tunnel.

Explanation:

Traffic originating from the Staff VLAN (192.168.20.0/24) is allowed to pass through the VPN tunnel interface (vpn_HQ-to-Branch_remote) toward the branch network (HQ-to-Branch_local). The service is set to **ALL**, meaning any protocol is permitted. NAT is disabled because VPN tunnels require **routing, not NAT**, to maintain end-to-end addressing. Security profiles (“default, certificate-inspection”) ensure encrypted traffic is validated.

Why it’s important:

This creates the **site-to-site connectivity** needed for HQ employees to access branch resources securely.

ID6: Staff To Servers

Purpose:

Allows devices in Staff VLAN (20) to access required services on Servers VLAN (40).

Explanation:

This rule permits staff to reach server resources using only the protocols needed:

FTP – File transfers.

MySQL – Database access

RADIUS – Authentication

NAT is enabled for traffic translation between VLANs. Basic security is applied using Antivirus and SSL Inspection, while other filters are disabled since the traffic is internal. Logging is enabled for all sessions.

Why it’s important:

Ensure staff can access essential server services while limiting unnecessary or unauthorized traffic, improving both security and control.

Name 	Staff_to_Servers
Incoming Interface	 Staff (Vlan 20) 
Outgoing Interface	 Servers (Vlan 40) 
Source	 all + 
Destination	 all + 
Schedule	 always 
Service	 FTP   MYSQL   RADIUS  +
Action	 ACCEPT  DENY  LEARN

Firewall / Network Options

NAT	☒
IP Pool Configuration	Use Outgoing Interface Address Use Dynamic IP Pool
Preserve Source Port	☐

Security Profiles

AntiVirus	☒	AV default 	
Web Filter	☐		
DNS Filter	☐		
Application Control	☐		
IPS	☐		
SSL Inspection	☒	SSL certificate-inspection 	

Logging Options

Log Allowed Traffic	☒	Security Events All Sessions
Generate Logs when Session Starts	☒	
Capture Packets	☒	

Comments  0/1023

Enable this policy ☒

ID7: Guests Internet

Purpose:

Allows devices in the Guests VLAN (Vlan 30) to access the internet through the WAN (port1) interface.

Explanation:

This firewall policy permits traffic originating from the Guests network to reach external destinations on the internet. The rule allows only the following essentials.

services:

- DNS – Required for resolving domain names (e.g., converting *google.com* to an IP address)
- HTTP – Standard web browsing
- HTTPS – Secure, encrypted web browsing
- PING – Allows basic connectivity testing.

Why it's important:

This policy provides controlled internet access to guest users while enforcing strict security inspection. Only necessary web-related services are allowed, minimizing exposure and preventing guests from using unauthorized or risky protocols. The applied security profiles help maintain network protection even for unmanaged guest devices.

Name ⓘ	Guests Internet	
Incoming Interface	Guests (Vlan 30) ▼	
Outgoing Interface	WAN (port1) ▼	
Source	all + ✕	
Destination	all + ✕	
Schedule	always ▼	
Service	DNS ✕ HTTP ✕ HTTPS ✕ PING ✕ +	
Action	☒ ACCEPT ☐ DENY ☐ LEARN	

Firewall / Network Options

NAT ☒

IP Pool Configuration ☒ Use Outgoing Interface Address ☐ Use Dynamic IP Pool

Preserve Source Port ☐

Security Profiles

AntiVirus	☒	AV default	✎
Web Filter	☒	WEB default	✎
DNS Filter	☒	DNS default	✎
Application Control	☒	APP default	✎
IPS	☒	IPS default	✎
SSL Inspection	☐	SSL certificate-inspection	✎

Logging Options

Log Allowed Traffic ☒ Security Events ☒ All Sessions

Generate Logs when Session Starts ☒

Capture Packets ☒

Comments 0/1023

Enable this policy ☒

SNAT and DNAT

The session table output confirms that the FortiGate firewall is actively processing traffic across all VLANs. The presence of multiple ICMP sessions verifies successful connectivity testing between Staff, Guest, and Server networks, proving that inter-VLAN routing and firewall policies are functioning correctly. DHCP broadcast traffic (UDP ports 67/68) indicates that endpoint devices are successfully obtaining IP addresses from the FortiGate DHCP server. Additionally, the UDP/500 session confirms that IPsec VPN negotiation with the Branch network is active. Overall, the session list validates that the network is fully operational and that all configured policies and services are working as intended.

```
HQ-Firewall # get system session list
PROTO  EXPIRE  SOURCE          SOURCE-NAT      DESTINATION      DESTINATION-NAT
udp     178     0.0.0.0:68      -               255.255.255.255:67 -
icmp    42      192.168.40.1:49334 -               192.168.40.2:8   -
icmp    6       192.168.20.1:57420 -               192.168.20.3:8   -
icmp    23      192.168.30.1:41129 -               192.168.30.2:8   -
icmp    57      192.168.40.1:41151 -               192.168.40.4:8   -
icmp    17      192.168.20.1:49317 -               192.168.20.4:8   -
icmp    51      192.168.40.1:20667 -               192.168.40.3:8   -
icmp    36      192.168.30.1:28850 -               192.168.30.4:8   -
icmp    29      192.168.30.1:8366  -               192.168.30.3:8   -
icmp    0       192.168.20.1:16455 -               192.168.20.2:8   -
udp     60      192.168.72.100:500 -               192.168.72.200:500 -
```

VPN

The firewall shows a connection between **192.168.72.100** and **192.168.72.200** using **UDP port 500**, which is the standard port for VPNs. This means these two devices are setting up or maintaining a **VPN tunnel** to securely communicate.

```
HQ-Firewall # get system session list
PROTO  EXPIRE  SOURCE          SOURCE-NAT      DESTINATION      DESTINATION-NAT
udp     28      0.0.0.0:68      -               255.255.255.255:67 -
icmp    53      192.167.20.2:21602 -               192.168.20.2:8   -
icmp    52      192.167.20.2:21346 -               192.168.20.2:8   -
icmp    51      192.167.20.2:21090 -               192.168.20.2:8   -
icmp    50      192.167.20.2:20834 -               192.168.20.2:8   -
udp     168     192.168.72.100:500 -               192.168.72.200:500 -

HQ-Firewall #
```

Output testing

FortiGate VM64-KVM HQ-Firewall

Dashboard

Security Fabric

FortiView

Network

System

Policy & Objects

Security Profiles

VPN

User & Device

Log & Report

Forward Traffic

Local Traffic

Sniffer Traffic

System Events

Router Events

VPN Events

User Events

Endpoint Events

HA Events

Security Rating Events

WAN Opt. & Cache Events

AntiVirus

Web Filter

DNS Query

Application Control

Intrusion Prevention

Anomaly

Learning Report

Log Settings

Threat Weight

Email Alert Settings

Monitor

Add Filter

#	Date/Time	Level	Action	Status	Message	VPN Tunnel
1	10 hours ago		tunnel-stats		IPsec tunnel statistics	HQ-to-Branch
2	10 hours ago		tunnel-stats		IPsec tunnel statistics	HQ-to-Branch
3	10 hours ago		tunnel-stats		IPsec tunnel statistics	HQ-to-Branch
4	10 hours ago		tunnel-stats		IPsec tunnel statistics	HQ-to-Branch
5	10 hours ago		tunnel-stats		IPsec tunnel statistics	HQ-to-Branch
6	10 hours ago		tunnel-stats		IPsec tunnel statistics	HQ-to-Branch
7	11 hours ago		tunnel-stats		IPsec tunnel statistics	HQ-to-Branch
8	11 hours ago		negotiate	success	negotiate IPsec phase 2	HQ-to-Branch
9	11 hours ago		negotiate	success	progress IPsec phase 2	HQ-to-Branch
10	11 hours ago		negotiate	success	progress IPsec phase 2	HQ-to-Branch
11	11 hours ago		phase2-up		IPsec phase 2 status change	HQ-to-Branch
12	11 hours ago		install_sa		install IPsec SA	HQ-to-Branch
13	11 hours ago		phase2-down		IPsec phase 2 status change	HQ-to-Branch
14	11 hours ago		negotiate	success	negotiate IPsec phase 2	HQ-to-Branch
15	11 hours ago		negotiate	success	progress IPsec phase 2	HQ-to-Branch
16	11 hours ago		negotiate	success	progress IPsec phase 2	HQ-to-Branch
17	11 hours ago		phase2-up		IPsec phase 2 status change	HQ-to-Branch
18	11 hours ago		install_sa		install IPsec SA	HQ-to-Branch
19	11 hours ago		phase2-down		IPsec phase 2 status change	HQ-to-Branch
20	11 hours ago		negotiate	success	negotiate IPsec phase 2	HQ-to-Branch
21	11 hours ago		negotiate	success	progress IPsec phase 2	HQ-to-Branch
22	11 hours ago		negotiate	success	progress IPsec phase 2	HQ-to-Branch
23	11 hours ago		tunnel-up		IPsec connection status change	HQ-to-Branch
24	11 hours ago		phase2-up		IPsec phase 2 status change	HQ-to-Branch
25	11 hours ago		install_sa		install IPsec SA	HQ-to-Branch
26	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch
27	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch
28	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch
29	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch
30	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch
31	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch
32	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch
33	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch
34	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch
35	11 hours ago		negotiate	success	progress IPsec phase 1	HQ-to-Branch

Firewall

Refresh

Reset Statistics

Bring Up

Bring Down

Name

Type

Remote Gateway

User Name

Incoming Data

Outgoing Data

Phase 1

Phase 2 Selectors

HQ-to-Branch

Site to Site - FortiGate

192.168.72.100

3.94 kB

2.46 kB

HQ-to-Branch

HQ-to-Branch

HQ-Firewall

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The Guest2 client initially attempted DHCP discovery but did not receive an offer, which is common when the interface or DHCP service is still initializing. On the second attempt, the device successfully completed the full DORA process (Discover, Offer, Request, Acknowledge), receiving the IP address 192.168.30.2/24 from the DHCP server running on the FortiGate VLAN 30 interface (192.168.30.1). The client also received the correct gateway, DNS server, and a 7-day lease, confirming that the Guest VLAN DHCP configuration is working correctly and that Guest2 is fully integrated into the network.

```
Guest2> ip dhcp
DDD
Can't find dhcp server

Guest2> ip dhcp
DDORA IP 192.168.30.2/24 GW 192.168.30.1

Guest2> show ip

NAME           : Guest2[1]
IP/MASK        : 192.168.30.2/24
GATEWAY        : 192.168.30.1
DNS            : 8.8.8.8
DHCP SERVER    : 192.168.30.1
DHCP LEASE     : 604781, 604800/302400/529200
MAC            : 00:50:79:66:68:05
LPORT          : 20078
RHOST:PORT     : 127.0.0.1:20079
MTU            : 1500
```

```
Guest2> ping 192.168.10.3
```

```
192.168.10.3 icmp_seq=1 timeout  
192.168.10.3 icmp_seq=2 timeout  
192.168.10.3 icmp_seq=3 timeout  
192.168.10.3 icmp_seq=4 timeout  
192.168.10.3 icmp_seq=5 timeout
```

```
Guest2> ping 192.168.72.200
```

```
192.168.72.200 icmp_seq=1 timeout  
192.168.72.200 icmp_seq=2 timeout  
192.168.72.200 icmp_seq=3 timeout  
192.168.72.200 icmp_seq=4 timeout  
192.168.72.200 icmp_seq=5 timeout
```

```
Guest2> ping 192.167.20.3
```

```
192.167.20.3 icmp_seq=1 timeout  
192.167.20.3 icmp_seq=2 timeout  
192.167.20.3 icmp_seq=3 timeout  
192.167.20.3 icmp_seq=4 timeout  
192.167.20.3 icmp_seq=5 timeout
```

```
Raduis> ip dhcp  
DDORA IP 192.168.40.2/24 GW 192.168.40.1
```

```
Raduis> show ip
```

```
NAME           : Raduis[1]  
IP/MASK         : 192.168.40.2/24  
GATEWAY         : 192.168.40.1  
DNS             : 8.8.8.8  
DHCP SERVER     : 192.168.40.1  
DHCP LEASE      : 604792, 604800/302400/529200  
MAC             : 00:50:79:66:68:08  
LPORT           : 20090  
RHOST:PORT      : 127.0.0.1:20091  
MTU             : 1500
```

```
Branch_Pc> ping 192.168.20.3
```

```
84 bytes from 192.168.20.3 icmp_seq=1 ttl=62 time=3.753 ms  
84 bytes from 192.168.20.3 icmp_seq=2 ttl=62 time=2.906 ms  
84 bytes from 192.168.20.3 icmp_seq=3 ttl=62 time=2.578 ms  
84 bytes from 192.168.20.3 icmp_seq=4 ttl=62 time=2.809 ms  
84 bytes from 192.168.20.3 icmp_seq=5 ttl=62 time=2.693 ms
```

```
Branch_Pc> █
```